Unit - 04

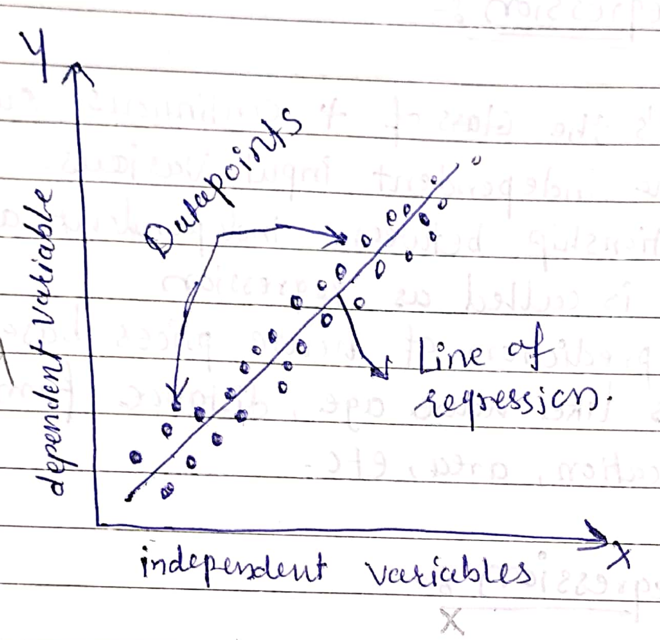
~~Linear~~ Regression :-

- It predict's the ~~Class of~~ + continuous output variables based on the independent input various.
- The Relationship between independent and dependent variable is called as regression
- like the prediction of house prices based on different parameters like house age, distance from the main road, location, area, etc.

* Linear Regression :-

- Linear regression is a type of Supervised machine Learning algorithm that computes the linear relationship between a dependent variable and one or more independent features.
- when the number of the independent feature, is 1 then it is known as Univariate Linear regression, and in the case of more than one feature, it is known as multivariate linear regression

- The goal of the algorithm is to find the best linear equation that can predict the value of the dependent variable based on the independent variable.
- The equation provides a straight line that represents the relationship between the dependent and independent variable.
- The slope of the line indicates how much the dependent variable changes for a unit change in the independent variable.



- Linear regression is used in many different fields, including finance, economics, and psychology, to understand and predict the behavior of a particular variable.
- Mathematically, we can represent a linear regression as:

$$y = a_0 + a_1 x + \varepsilon$$

Here,
Y = Dependent Variable (target variable)
X = Independent Variable (predictor Variable)
a_0 = intercept of the line
a_1 = Linear regression coefficient
ε = random error.

* Types of Linear Regression

Linear regression can be further divided into two types of the algorithm:
① Simple Linear Regression
② Multiple Linear Regression.

① Simple Linear Regression :-

If a Single independent variable is used to predict the value of a numerical dependent variable, then such a Linear regression algorithm is called simple Linear Regression
- Mathematically represented as -

$$Y = \alpha_0 + \alpha_1 x_1$$
$$Y = C + mX$$

Where, Y = dependent Variable
X = independent variable
α_0, α_1 = Coefficient of regression.

(2) Multiple Linear Regression :-

- If more than one independent variable is used to predict the value of a numerical dependent variable, then such a linear regression algorithm is called Multiple Linear Regression.
- Mathematically represented as

$$Y = \alpha_0 + \alpha_1 x_1 + \alpha_2 X_2 + \alpha_3 X_3 + \cdots + \alpha_m X_m$$

Where,

α_i = Regression coefficient

X_i = Indepenent variable

Y = ~~Ind~~ Dependent variable.